Pneumonia

By

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- **Pneumonia** is inflammation of the lung parenchyma leading to consolidation of part / the whole lung due to **intra-alveolar exudation.**

- Pneumonia is the single **largest infectious cause** of death in children worldwide. For US adults, pneumonia is the **most common cause of hospital admissions** other than women giving birth.

- **Older people** have higher risk of getting pneumonia, and are more likely to die from it if they do.

- Pneumonia is **the most common cause of sepsis and septic shock,** causing 50% of all episodes.

Classification

Aetiological

- Bacterial
- Atypical
- Viral
- Fungal
- Parasitic
- Autoimmune
- Radiation, chemical

Empiricist's classification

Community acquired pneumonia (CAP)

-Str. Pneumoniae
-H.Influenza
-M. pneumoniae

Hospital acquired pneumonia (HAP)

-Gm. –ve (Kl. P. aeraginosa, E.coli)
-Staph. aureus
-Strept.,pneumoniae
-Anaerobes

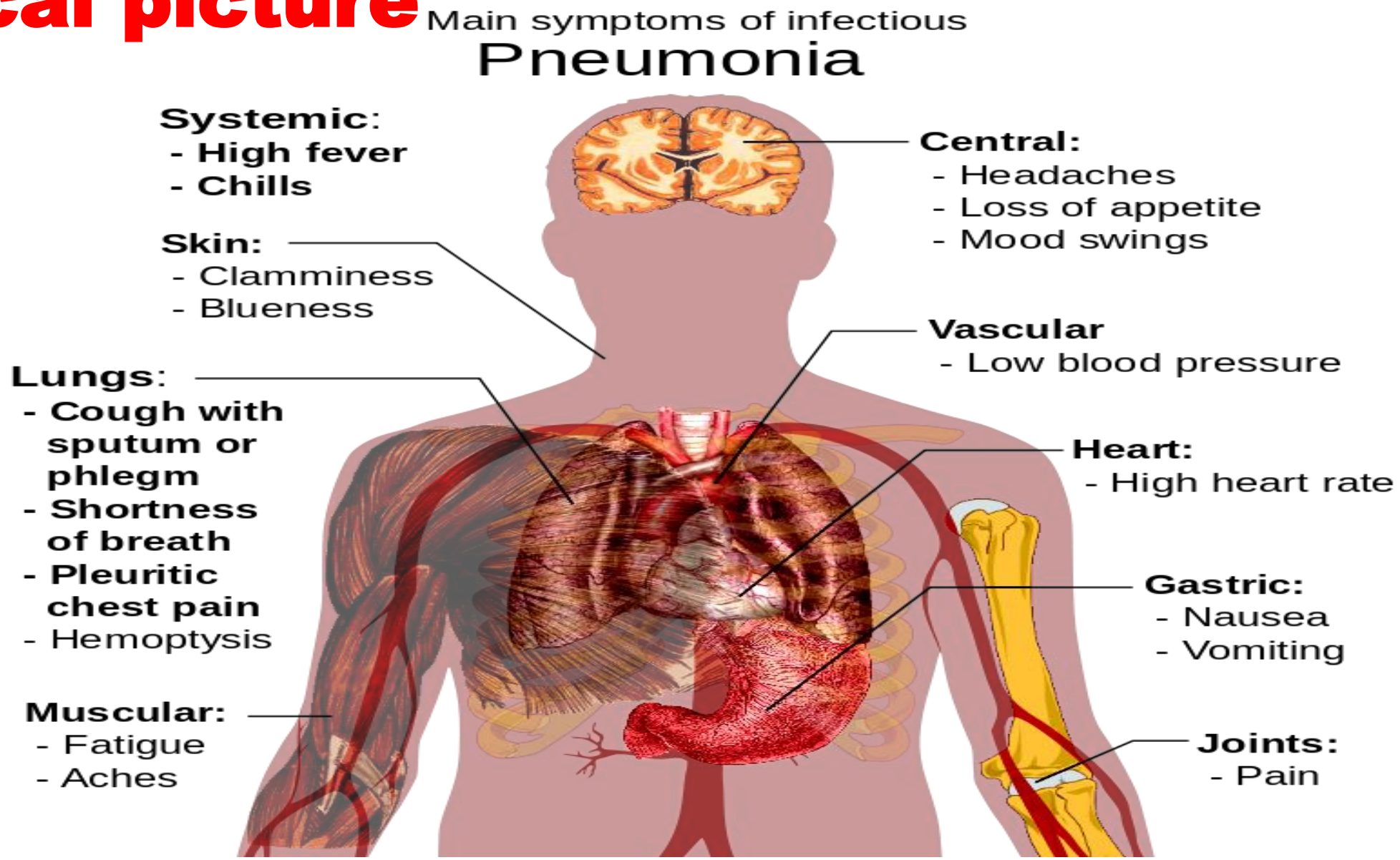
Pneumonia (immunocom-promised)

-Pneumocystis.
-Fungal
-Atypial mycobacteria
-CMV

Anatomical

- Lobar
- Lobular bronchopneumonia
- Segmental
- Interstitial

Clinical picture



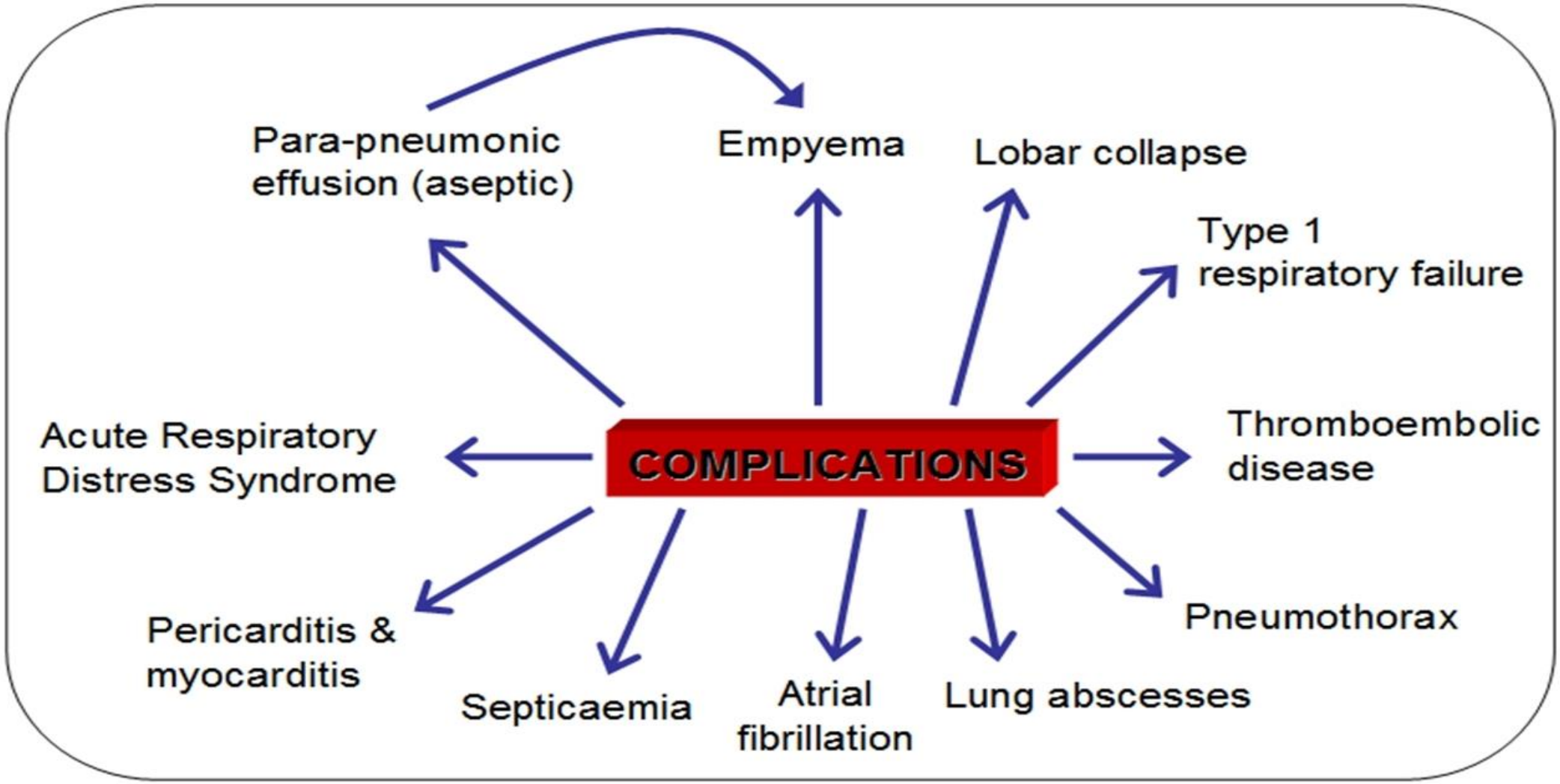
Diagnosis

- ❑ Clinical picture
- ❑ Radiology
- ❑ Cultures (sputum, blood..)
- ❑ Others (CBC- ESR-CPR)





Complications



Treatment

Antibiotics

Pharmacokinetics

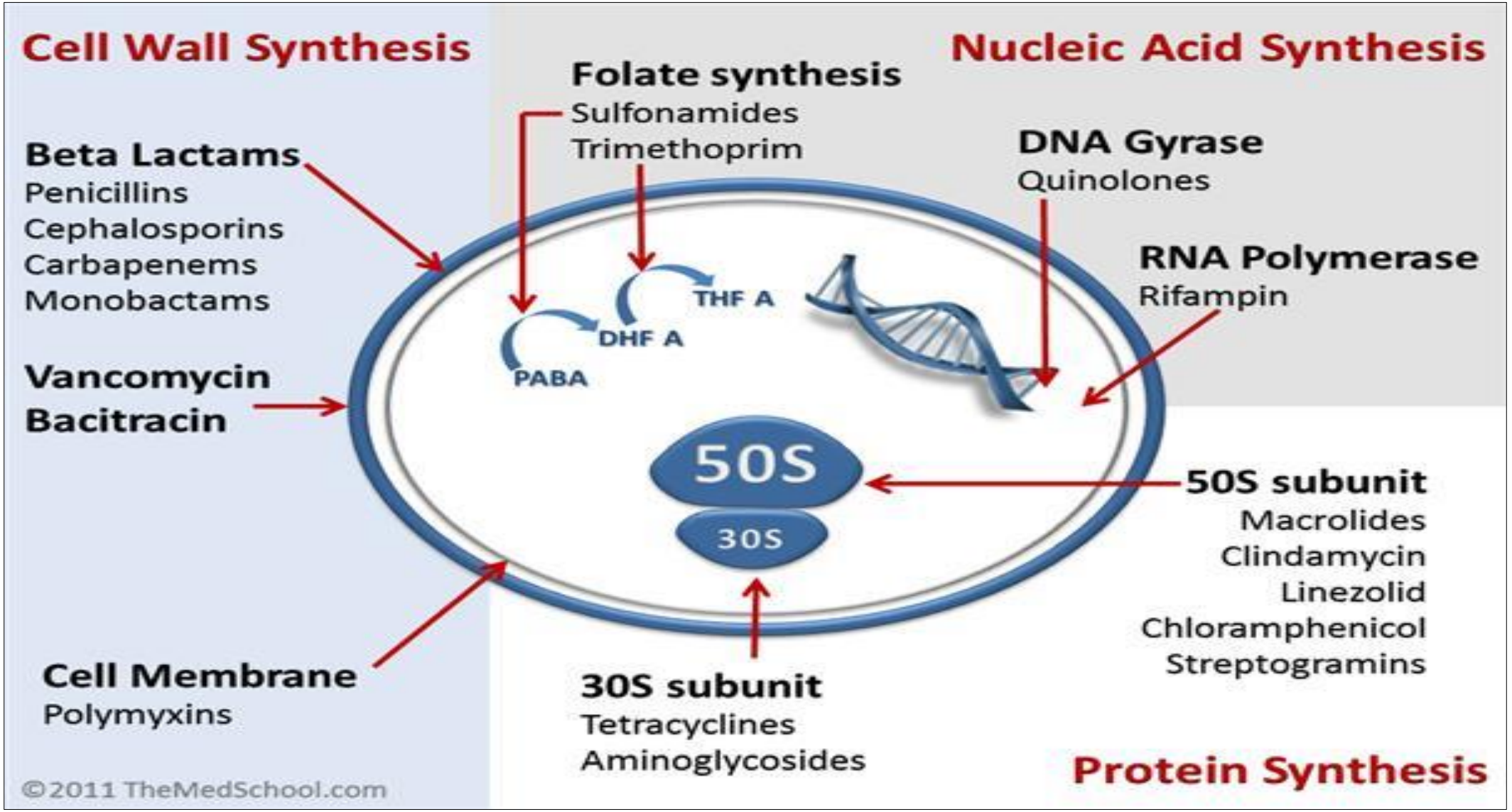
Liver metabolism

Amoxicillin/clavulanic
Pipracillin
Carpabenems
Linezolid
Macrolides

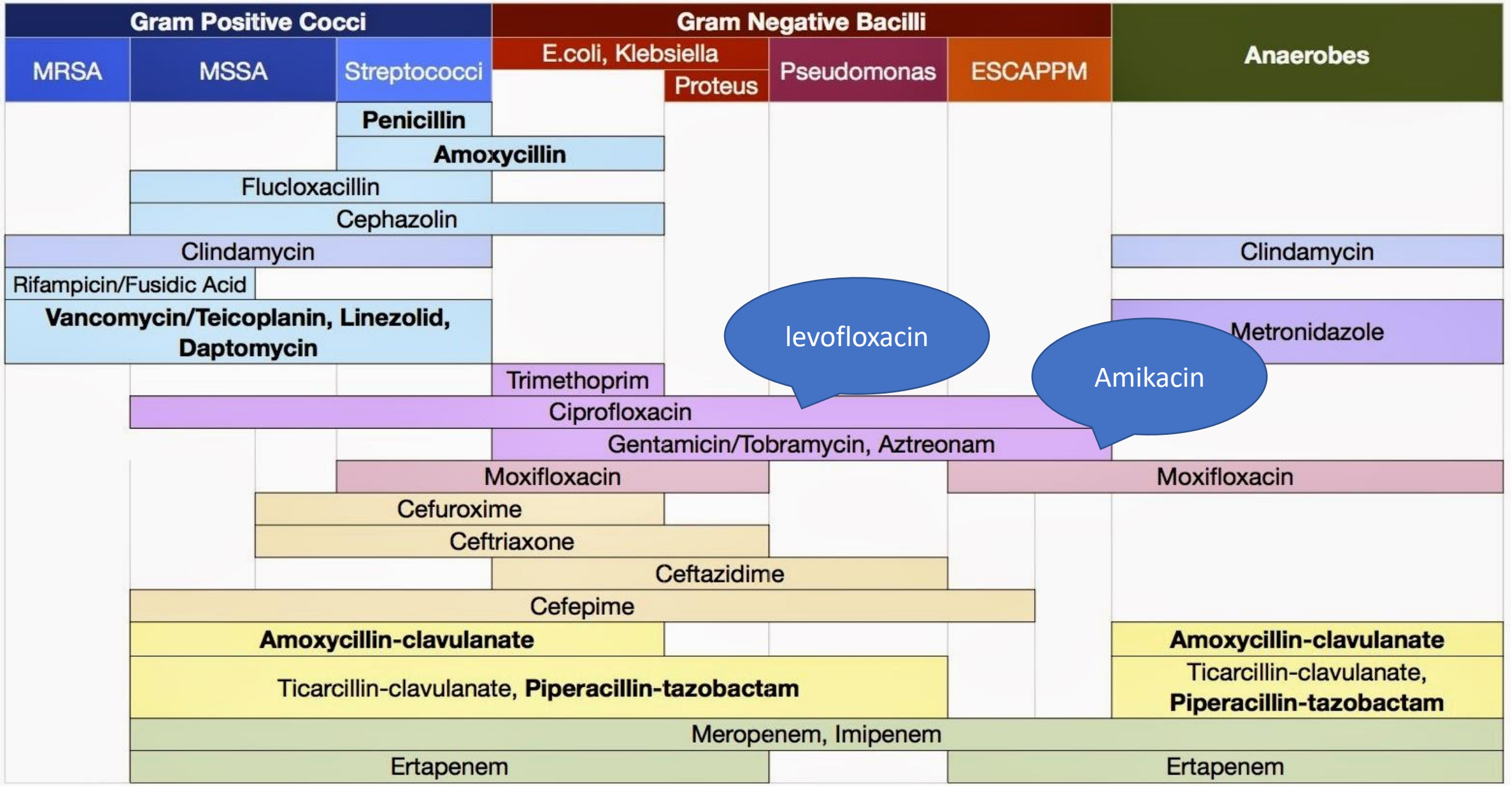
Elimination by kidney

Aminoglycosides
Quinolones
Cephalosporins
Glycopeptides

Mechanism of action



Specrtum of Antibacterial Drugs



Side effects

Penicillins, Cephalosporins
Rash, allergic reactions, GIT upset

Quinolones
GIT upset, dysglycaemia
Nephrotoxicity fits, arrhythmia

Carpabenems
Diarrhea, nausea, vomiting, hepatotoxicity

Aminoglycosides
Nephrotoxicity, Ototoxicity

Macrolides
Vomiting, diarrhea

Clindamycin
Pseudomembranous colitis, B.M suppression

Linezolid
B.M. suppression, GIT upset

Glycopeptides
Nephrotoxicity, B.M. suppression

CAP Treatment: 2007 IDSA/ATS Guidelines

Outpatient	General Medical Ward	ICU/Severe
If no significant risks for DRSP*: **Macrolide or doxycycline** If risks for DRSP*: **Antipneumococcal fluoroquinolone** **OR** High-dose amoxicillin (3 gm/day) or high dose amoxicillin/clavulanate (4 gm/day) **plus** macrolide (if amoxicillin is used and there is a concern for H. influenzae, use macrolide active for -lactamase producing strains)	**Beta-lactam** (ceftriaxone, cefotaxime, ampicillin/sulbactam, ertapenem) **plus macrolide** (can use doxycycline if macrolide not tolerated) **OR** **Antipneumococcal fluoroquinolone alone**	**Beta-lactam** (ceftriaxone, cefotaxime, ampicillin/sulbactam) **plus IV azithromycin or IV fluoroquinolone** If concern for Pseudomonas (eg, presence of structural lung disease such as bronchiectasis): antipseudomonal agent (piperacillin/tazobactam, imipenem, meropenem, or cefepime) **plus** antipseudomonal fluoroquinolone (ciprofloxacin or high dose levofloxacin); If concern for MRSA: add vancomycin or linezolid

***RF for DRSP: >65, exposure to children in day care, alcoholism or other severe underlying disease, or recent antibiotics**

Table 4. Recommended Initial Empiric Antibiotic Therapy for Hospital-Acquired Pneumonia (Non-Ventilator-Associated Pneumonia)

Not at High Risk of Mortality[a] and no Factors Increasing the Likelihood of MRSA[b,c]	Not at High Risk of Mortality[a] but With Factors Increasing the Likelihood of MRSA[b,c]	High Risk of Mortality or Receipt of Intravenous Antibiotics During the Prior 90 d[a,c]
One of the following:	One of the following:	Two of the following, avoid 2 β-lactams:
Piperacillin-tazobactam[d] 4.5 g IV q6h	Piperacillin-tazobactam[d] 4.5 g IV q6h	Piperacillin-tazobactam[d] 4.5 g IV q6h
OR	OR	OR
Cefepime[d] 2 g IV q8h	Cefepime[d] or ceftazidime[d] 2 g IV q8h	Cefepime[d] or ceftazidime[d] 2 g IV q8h
OR	OR	OR
Levofloxacin 750 mg IV daily	Levofloxacin 750 mg IV daily	Levofloxacin 750 mg IV daily
	Ciprofloxacin 400 mg IV q8h	Ciprofloxacin 400 mg IV q8h
	OR	OR
Imipenem[d] 500 mg IV q6h	Imipenem[d] 500 mg IV q6h	Imipenem[d] 500 mg IV q6h
Meropenem[d] 1 g IV q8h	Meropenem[d] 1 g IV q8h	Meropenem[d] 1 g IV q8h
	OR	OR
	Aztreonam 2 g IV q8h	Amikacin 15–20 mg/kg IV daily
		Gentamicin 5–7 mg/kg IV daily
		Tobramycin 5–7 mg/kg IV daily
		OR
		Aztreonam[a] 2 g IV q8h
	Plus:	Plus:
	Vancomycin 15 mg/kg IV q8–12h with goal to target 15–20 mg/mL trough level (consider a loading dose of 25–30 mg/kg × 1 for severe illness)	Vancomycin 15 mg/kg IV q8–12h with goal to target 15–20 mg/mL trough level (consider a loading dose of 25–30 mg/kg IV × 1 for severe illness)
	OR	OR
	Linezolid 600 mg IV q12h	Linezolid 600 mg IV q12h
		If MRSA coverage is not going to be used, include coverage for MSSA. Options include: Piperacillin-tazobactam, cefepime, levofloxacin, imipenem, meropenem. Oxacillin, nafcillin, and cefazolin are preferred for the treatment of proven MSSA, but would ordinarily not be used in an empiric regimen for HAP.
If patient has severe penicillin allergy and aztreonam is going to be used instead of any β-lactam–based antibiotic, include coverage for MSSA.		

Table 3. Suggested Empiric Treatment Options for Clinically Suspected Ventilator-Associated Pneumonia in Units Where Empiric Methicillin-Resistant *Staphylococcus aureus* Coverage and Double Antipseudomonal/Gram-Negative Coverage Are Appropriate

A. Gram-Positive Antibiotics With MRSA Activity	B. Gram-Negative Antibiotics With Antipseudomonal Activity: β-Lactam–Based Agents	C. Gram-Negative Antibiotics With Antipseudomonal Activity: Non-β-Lactam–Based Agents
Glycopeptides[a] Vancomycin 15 mg/kg IV q8–12h (consider a loading dose of 25–30 mg/kg × 1 for severe illness)	Antipseudomonal penicillins[b] Piperacillin-tazobactam 4.5 g IV q6h[b]	Fluoroquinolones Ciprofloxacin 400 mg IV q8h Levofloxacin 750 mg IV q24h
OR	OR	OR
Oxazolidinones Linezolid 600 mg IV q12h	Cephalosporins[b] Cefepime 2 g IV q8h Ceftazidime 2 g IV q8h	Aminoglycosides[a,c] Amikacin 15–20 mg/kg IV q24h Gentamicin 5–7 mg/kg IV q24h Tobramycin 5–7 mg/kg IV q24h
	OR	OR
	Carbapenems[b] Imipenem 500 mg IV q6h[d] Meropenem 1 g IV q8h	Polymyxins[a,e] Colistin 5 mg/kg IV × 1 (loading dose) followed by 2.5 mg × (1.5 × CrCl + 30) IV q12h (maintenance dose) [135] Polymyxin B 2.5–3.0 mg/kg/d divided in 2 daily IV doses
	OR	
	Monobactams[f] Aztreonam 2 g IV q8h	

Choose one gram-positive option from column A, one gram-negative option from column B, and one gram-negative option from column C. Note that the initial doses suggested in this table may need to be modified for patients with hepatic or renal dysfunction.

Antibiotics Drug Interactions

Antibiotic	Interacting drug	Mechanism
Amoxicillin-clavulaninic	BCG, Cholera, Typhoid vaccines*	Inactivation of these vaccines
	Aspirin	Both increase serum level of the other
Cephalosporins(cefip-ime,ceftazidime)	Live att. Vaccines*	Inactivation of vaccines
	Warfarin	↓ vit. K forming GIT flora→↑ warfarin action
Piperacillin-Tazobactum	Heparin, warfarin*	Promote their toxicity (plt. Aggregation inhibitor)
	Macrolides	Its bactericidal effect contradict their bacteriostatic action
Carpabenems (imipenem, meropenem)	Live att. Vaccines*	Inactivation of vaccines
	Digoxin	Altering gut flora→ increase in digoxin level→toxicity
	Theophylline	Increased risk of seizures (imipenem)

Antibiotics Drug Interactions

Antibiotic	Interacting drug	Mechanism
Quinolones	Iron supplement*	Decrease their GIT absorbtion→↓ action
	Amiodarone	↑ QT interval → arrythmias
	Digoxin	↑ digoxin level due to alteration in GIT flora
	Macrolides, azoles	↑ QT interval → arrythmias
	Oral hypoglycaemic	Synergism→ dysglycaemia
	Steroids	Increase risk of tendinitis, tendon rupture
	warfarin	Decreased warfarin metabolism→ increased level and toxicity
Amikacin	Amphotericin	Both nephrotoxic and ototoxic
	Loop diuretics	Synergism (more nephrotoxicity and ototoxicity)
	Teicoplanin	
	Muscle relaxants	Synergism→ more muscle weakness
	Warfarin	Amikin increase warfarin level by unknown mechanism

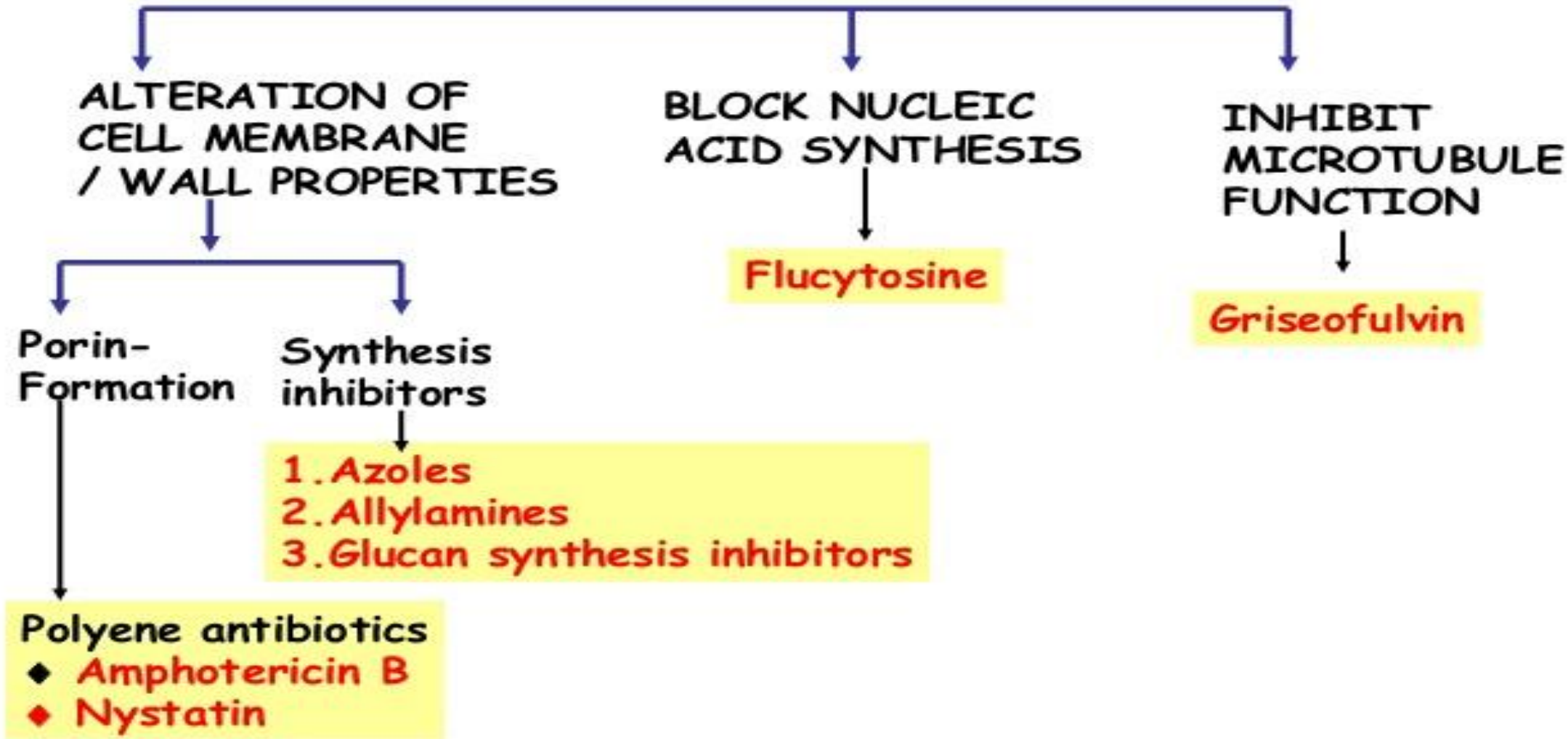
Antibiotics Drug Interactions

Antibiotic	Interacting drug	Mechanism
Linezolid	Dopamine, dobutamine, norepinephrine*	Synergism. May precipitate HTN crisis
	Serotononergic drugs*	Linezolid inhibit MAO→ marked ↑ in serotonin level→ CNS toxicity
	L-dopa, INH, lithium	CNS toxicity (increased serotonin level)
	B 2 agonists	Synergism. May precipitate HTN crisis
Teicoplanin	Aminoglycoside *	Incompatible combination (don't use the same IV line)
	Other nephrotoxic drugs	Nephrotoxic
Vancomycin	Aminoglycosides	Increased nephrotoxicity
	Live att. Vaccines*	Inactivation

Antibiotics Drug Interactions

Antibiotic	Interacting drug	Mechanism
Macrolides	Azole antifungals (fluconazole)*	↑ QT interval → arrythmias
	Levofloxacin	↑ QT interval → arrythmias
	Warfarin, steroid*	Their metabolism↓ by macrolides→↑ level
	Rifampicin, phenytoin*	These induce metabolism of macrolides→ ↓ effect
	Amiodarone*	Its metabolism↓ by macrolides→↑ level
	Digoxin	Change in GIT flora→ ↑ digoxin level and toxicity
	Clopidogrel	Decreased action dt. Decreased conversion to active meatbolites
	Piperacillin	Its bactericidal effect contradict macrolides' bacteriostatic action
	Oral hypoglycemic/insulin	Synergism → dysglycaemia

ANTIFUNGALS



Antifungal Drugs: Azole antifungals

Mechanism: inhibit ergosterol synthesis→ **cell wall disruption**

Spectrum: some yeasts, dimorphic fungi, dermatophytes, asperigellus.

Side effects: nausea, vomiting, hepatotoxicity

Interactions:

- **Absorbtion** affected by H2 blockers.
- They are considered as **hepatic enzyme inhibitors**→ ppt. toxicity of other drugs. e.g. warfarin, quinolones, antituberculous
- **Rifampicin** is a strong hepatic enzyme inducer-→ decrease action of these drugs.

Antifungal Drugs:Echinocandins Micafungin

Mechanism: glucan synthesis inhibitors (cell wall)

Spectrum: candidiasis

Dose 100 mg/day for 10-15 days.

Side effects: minor GIT upset, headache, flushing, rash, incressed liver enzymes, postmarketing Reports (Blood and lymphatic system disorders: white blood cell count decreased, hemolytic anemia, disseminated intravascular coagulation)

Interactions: very few interactions clinical significance

Antifungal Drugs: Echinocandins caposfungin

Mechanism: glucan synthesis inhibitors (cell wall)

Spectrum: candidiasis, aspergillosis

Dose: loading 70mg IV infused over 1hour

Maintenance 50 mg qDay IV infused over 1hour 10-14 days

Side effects: GIT upset, incressed liver enzymes (more than micafungin), respiratory failure, headache, flushing, rash postmarketing Reports (Blood and lymphatic system disorders: white blood cell count decreased, hemolytic anemia, disseminated intravascular coagulation)

Interactions: rifampicin, phenytoin, dexamethasone induce its metabolism→decreased action

Antiviral: Tamiflu (oseltamivir)

Dose:

- H1N1 prophylaxis : 75 mg oral once/day (5days). within 7 days of exposure
- H1N1 treatment: 75 mg oral twice/day (5days). within 48 hrs of exposure

Side effects: nausea, vomiting, rash, hepatitis, arrhythmia

Interactions: influenza vaccines (decreased effect of vaccine by antagonism)

Interactive cases

Case 1

- A 25 year old man presented to the clinic with a 5 days-history of fever, cough, expectoration of yellowish sputum and rt. Sided chest pain. He looks apparently healthy with no signs of respiratory distress or cynosis with normal vital signs. CXR was done and showed RT. Lobar lobe pneumonia.

- **Q1:** What's the most likely organism?

- **Q2:** Which antibiotic(s) should be prescribed by the physician?

- **Q3:** The patient mentioned recent use of antibiotic. Which drug is better to be prescribed?

- A case of **community acquired pneunmonia**, mostly **mild** case in otherwise healthy patient

- **A1:** Strept. Pneumoniae, H. influenza, Chlamydia.

- **A2:** Macrolides (e.g. clarithromycin 500 mg/12 hour)

- **A3:** Recent exposure to antibiotics increase risk of drug resistant organism so different agents are better to be used:

❑Antipneumococcal quinolone (levofloxacin 750 mg/day)

❑High dose amoxicillin /clavulanic acid 4gm/day plus macrolide

❑ High dose amoxicillin 3gm/day.

Case 2

- A 58 year old diabetic woman was admitted to hospital for control of her blood sugar and debridement of her diabetic foot. On the 4th day, she developed fever, tachypnea, worsening oxygenation. Urgent CXR revealed bronchopneumonia. She became hypotensive and was admitted to the ICU to persue treatment and monitoring.

- **Q1:** What organisms may be implicated?
- **Q2:** Which antibiotic(s) is/are to be prescribed in such a case?

- This is a case of **severe hospital acquired pneumonia** with high risk of death.

- **A1:** -Gram –ve (pseudomonas, Kl., E.Coli)
  - Staph. Aureus.
  -Strept. Pneumoniae.
  -Anaerobes.
- **A2: Two antipseudomonal drugs plus antistaphylococcal drug** e.g.:
  - Cefipime 2 gm/8 hrs
  -levofloxacin 750 mg/day.
  -Linezolid 600 mg / 12 hrs

  } **IV**

Case 3

The following are the lab. Investigations of the previous patient:

CBC	Value
WBCS	22,000 H
Hb	11
Hct	37 %
Plt.	198,000

Renal Function	Value
Na	132
K	3.5 L

Electrolyte	Value
Urea	70 mg/dl H
Creatinine	2.5 mg/dl H

LFTs	Value
Bil	0.1 mg/dl
ALT	32 IU
AST	35IU
ALB.	3.6 mg/dl
TP	8.2 mg/dl

- **Q1:** The physician wants to change to (meropenem, amikacin, teicoplanin) as the patient is not responding. Report your considerations regarding the current regimen and the proposed one.

- **Q2:** Mention any further investigations you would like to request.

- **A1:** There's evident **renal impairment** (raised urea and creatinine)
- ***The current regimen:*** levofloxacin is relatively nephrotoxic and cefepime is excreted by the kidney. So, the **dose** of both must be **adjusted** if to be continued. Linezolid does not need dose adjustment.
- ***The proposed regimen:*** **Amikacin** is mainly excreted unchanged in urine, highly nephrotoxic so, it's **better** to be **avoided**. Also meropenem and teicoplanin are nephrotoxic so, extreme caution must be paid if given together and certainly **dose adjustment** is necessary.

- **A2: Creatinine clearance** to help dose adjustment.

  Blood sugar level monitoring (levofloxacin causes dysglycaemia)

Case 4

- A 60 year old man has community acquired pneumonia. He is maintained on ACEIs, clopidogril, spironolactone for treatment of IHD and amiodarone tablet for A.F. The patient was prescribed erythromycin 500 mg tab once/day for 5 days. After 2 days, he was rushed to the E.R. for palpitation where ECG monitoring revealed multiple ectopics.

- **Q1:** What explains these ECG abnormalities?
- **Q2:** The patient reported severe chest pain, his cardiac enzymes were elevated. He was diagnosed with new MI. What is the possible mechanism?

- **A1:** Either:
- Erythromy caused prolonged QT interval which lead to arrhythmias.
- Erythromycin decreased metabolism of amiodarone→↑ its level-→ toxicity.

- **A2:** Erythromycin decreased metabolism of clopidogel into its active metabolites-→ decreased effect→ thrombosis of coronary arteries→ MI. This patient should have had his dose increased.


THANK YOU
ANTI-
BIOTIC